



ÇUKUROVA UNIVERSITY
FACULTY OF ENGINEERING



DEPARTMENT OF COMPUTER ENGINEERING

GRADUATION THESIS

FINAL REPORT

**REAL-TIME PARKING LOT OCCUPANCY DETECTION
SYSTEM**

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THESIS ABSTRACT

This project presents a comprehensive, real-time parking lot occupancy detection system utilizing a hybrid computer vision approach. Finding available parking spaces in urban environments is a significant challenge, and traditional sensor-based systems often lack scalability. The proposed system addresses this issue by combining the Ultralytics YOLOv8 object detection architecture with a custom-trained Convolutional Neural Network (CNN) based on the MobileNetV2 architecture.

Within the scope of the study, the CNRPark+EXT dataset was utilized to train the CNN model for binary classification of parking slots (empty/occupied), achieving high accuracy through data augmentation techniques. To maximize real-world reliability, a hybrid decision mechanism was developed. This mechanism evaluates CNN occupancy predictions alongside YOLOv8 bounding box overlaps, effectively mitigating common detection errors such as camera blind spots, occlusions, and shadows.

Furthermore, the system features a robust, web-based user interface developed entirely in Python using the Streamlit framework. The platform includes role-based access control (Admin and User functionalities), allowing administrators to dynamically draw custom Regions of Interest (ROIs) on newly uploaded video feeds and track user analysis logs via an integrated SQLite database. The expected outcome of this project is a scalable, cost-effective, and highly accurate smart parking management system that can be deployed in diverse environmental conditions without the need for physical sensor installations.

Keywords: Computer Vision, YOLO, OpenCV, Parking Detection, Streamlit

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ABBREVIATIONS AND SYMBOLS

CNN	Convolutional Neural Networks
CV	Computer Vision
GDPR	General Data Protection Regulation
IDE	Integrated Development Environment
VS Code	Visual Studio Code
YOLO	You Look Only Once

1. THESIS INTRODUCTION

With the increasing number of vehicles in urban environments, finding available parking spaces has become a significant problem. This situation causes traffic congestion, time loss, and increased fuel consumption in metropolitan areas. Smart parking systems supported by computer vision techniques provide an effective solution by monitoring parking areas automatically and informing users about available spaces in real time.

Compared to traditional hardware sensor-based parking detection systems, vision-based approaches offer a flexible and cost-effective alternative without requiring additional physical installation for each parking slot. However, vision-based systems often struggle with challenges such as dynamic lighting conditions, partial occlusions, and severe perspective distortions depending on the camera angle.

To overcome these challenges, this project proposes a novel deep learning-based hybrid object detection approach. Instead of relying solely on a single model, the system leverages the Ultralytics YOLOv8 architecture for vehicle localization and a custom MobileNetV2-based CNN model for patch-based slot classification. This dual-verification system ensures high precision.

The main objective of this study is to design, implement, and deploy a real-time parking occupancy detection system that is not only highly accurate but also user-friendly. To achieve this, a comprehensive web application has been developed. The application allows users to upload custom datasets, supports dynamic parking slot configuration through an interactive drawing canvas, and tracks historical occupancy analytics utilizing a relational database structure. This final report details the literature background, system architecture, hybrid detection methodologies, software implementation, and conclusive results of the project.

2. THESIS TOPIC

Previous studies on parking occupancy detection systems can generally be divided into sensor-based and vision-based approaches. While sensor-based systems provide high accuracy, they require substantial installation and ongoing maintenance costs. Therefore, vision-based systems utilizing surveillance cameras have become highly attractive in recent years due to their flexibility and scalability.

Recent research shows that state-of-the-art deep learning models such as YOLO achieve exceptional performance in real-time vehicle detection tasks. However, utilizing a standard YOLO model exclusively for parking lot management introduces several vulnerabilities. For instance, a vehicle passing through the aisles might be falsely identified as parked, or a parked vehicle might go undetected due to being blocked by a tree branch or cast shadows. Furthermore, purely classifying cropped parking slot images using a CNN can fail when vehicles overlap multiple slots.

The core topic of this thesis is the formulation and implementation of a hybrid algorithmic decision tree that resolves the deficiencies of standalone detection methods. The developed algorithm evaluates a specific parking slot by first applying a MobileNetV2 CNN model to cropped slot images to generate an initial probability score. Simultaneously, YOLOv8 scans the entire frame to map exact vehicle coordinates. The system then mathematically calculates the Intersection over Union (IoU) overlap ratio between the pre-defined slot coordinates and YOLO bounding boxes.

By establishing a rule-based logic—such as confirming occupancy if CNN confidence is overwhelmingly high (>99%) even if YOLO fails to detect due to blind spots, or rejecting occupancy if moving vehicles do not trigger the CNN slot threshold—the system maximizes precision. Additionally, the topic encompasses the software engineering challenge of transforming this complex algorithmic backend into an accessible, interactive, and multi-user web dashboard.

3. THESIS MATERIALS

This section describes the dataset, software frameworks, and development tools used in the implementation of the proposed parking occupancy detection system.

3.1 Software Stack and Frameworks

The technical architecture of the proposed real-time parking lot occupancy detection system is built upon a high-performance Python environment optimized for computer vision and relational data management. The core software components and deep learning frameworks utilized in this study are detailed below:

- **Ultralytics YOLOv8:** Adopted as the primary object localization framework to detect physical vehicle entities across full-scale frames. The model is explicitly constrained to evaluate specific target COCO indices: class 2 (car), class 3 (motorcycle), class 5 (bus), and class 7 (truck).
- **TensorFlow & Keras:** Utilized to compile, train, and execute a deep Convolutional Neural Network (CNN) tailored for patch-level binary classification (empty vs. occupied).
- **Streamlit Web Framework:** Used to construct the comprehensive, multi-user web application dashboard. It handles role-based UI rendering, dynamic asset caching via resource-caching annotations, and state monitoring throughout user sessions.
- **Streamlit Drawable Canvas (st_canvas) & PIL:** Embedded into the administrative control layer to provide an interactive, vector-based drawing interface over absolute image layouts, allowing dynamic geometric manipulation for custom region specification.
- **OpenCV (Open Source Computer Vision Library):** Deployed as the fundamental pixel-manipulation layer responsible for frame-by-frame video streaming, coordinate-based image cropping, geometric rendering, and RGB-BGR matrix conversions.
- **SQLite3 Engine:** Integrated as the local database abstraction layer to provide data persistence with row-factory mappings for transactional operations.

3.2 Dataset

To validate vehicle classification robustness under diverse outdoor deployments, the experimental phase of this study incorporates the benchmark CNRPark+EXT dataset . The dataset comprises thousands of parking slot images compiled from surveillance infrastructure under heavily challenging environmental permutations . The analytical characteristics and internal preprocessing pipelines developed for dataset preparation are outlined below:

- **Dataset Diversity:** The image directory holds structural representations of parking spaces captured across distinct weather conditions (e.g., sunny, overcast, rainy), extreme lighting variations, and sharp camera perspective distortions . These intentional variations ensure that the deep learning models develop high multi-environment generalization capabilities and maintain accurate detections despite cast shadows or partial vehicle occlusions .
- **Binary Labeling Schema:** The image metadata catalog follows a structured space-separated nomenclature (`sunny.txt`) where each specific image file path is explicitly mapped to a binary occupancy label: 0 representing an empty parking space and 1 representing an occupied parking space.
- **Data Validation Pipeline:** During initialization, the ingestion script systematically parses the raw text ledger, programmatically expands the absolute system paths, and applies validation cross-checks via operational file system indicators to automatically filter out any corrupted paths or missing image nodes.
- **Tensor Standardization:** Each isolated parking space patch is structurally resized to a rigid dimension of 150x150 pixels with 3-channel color representation to align perfectly with the target input layer of the compiled CNN network.
- **Real-Time Data Augmentation:** To enforce structural regularization and actively prevent deep learning overfitting during the optimization cycle, a robust data augmentation pipeline is executed via an image generator configuration. The input pixel matrices are normalized by rescaling values by a factor of $1/255$. Furthermore, the training frames are subjected to randomized transformations in real time, including horizontal flipping, a 15-degree maximum rotation boundary, a 10% structural width and height shift ratio, a 10% zoom magnification variance limit, and dynamic pixel brightness modifications restricted within a strict 0.7 to 1.3 scale range.

3.3 System Architecture

The structural layout of the developed smart parking platform establishes an end-to-end automated framework that transforms raw camera feeds into real-time analytical occupancy data . The architecture is designed across decoupled computational stages to optimize processing speed and maximize decision accuracy . The system encompasses five main processing stages :

1. **Image and Video Feed Acquisition:** The initialization layer continuously captures incoming visual assets from local directories, dataset catalogs (CNRPark+EXT), or active video streams .
2. **Coordinate Configuration and Scaling:** The system loads pre-defined parking slot coordinate profiles (CSVs) or interactive user annotations. Since surveillance frames are recorded at a high absolute resolution (2592×1944) but rendered on scaled monitors, the system automatically applies spatial resolution correction coefficients (x_{scale} and y_{scale}) to map bounding boxes perfectly onto the runtime pixel matrix.
3. **Automated Camera Identification:** If incoming assets lack explicit descriptive file labels, an automated spatial matching engine iterates through reference camera profiles (camera1 to camera9). It calculates the spatial overlap ratio between pre-configured slot boundaries and live YOLO vehicle detections. A match is verified if the intersection area divided by the slot area exceeds a threshold of 0.3, designating the profile with the highest confirmed matching ratio as the active host camera.
4. **Dual-Model Hybrid Inference:** This core computational layer executes two parallel neural network inferences: full-frame scanning using **Ultralytics YOLOv8** to localize vehicle entities, and patch-level classification using a **MobileNetV2 CNN** to generate occupancy probabilities for individual slot patches . A rigid decision tree resolves these outputs: a slot is marked as *Occupied* if both models confirm presence. If YOLO fails due to perspective distortion or occlusions, fallback rules trigger occupancy if the slot falls within the lower boundary blind-spot ($Y_{\text{start}} > \text{Frame}_{\text{height}} \times 0.70$) or if the CNN confidence clears an absolute override threshold (> 0.99).
5. **Database Logging and Visualization:** Computes aggregate parking metrics (total capacity, occupied/empty spaces, and percentage rates) and pipes them to an SQLite database for persistent logging . Concurrently, the visualization engine uses OpenCV to render red bounding box matrices over occupied slots and overlays identifier tags (`#SlotId`) onto the Streamlit dashboard .

4. POTENTIAL CONTRIBUTIONS TO THE INDUSTRY

The proposed hybrid parking occupancy detection system offers substantial technological and economic contributions to the intelligent transportation systems (ITS) and smart city industries. The primary industry contributions derived from the implementation of this study are as follows:

- **Cost-Effective Scalability:** By utilizing existing surveillance camera infrastructure and deep learning algorithms, the system eliminates the need for expensive physical hardware sensors (such as ultrasonic or magnetic ground sensors) at each individual parking slot. This significantly reduces both initial installation costs and long-term maintenance overhead, providing a highly scalable solution for large metropolitan parking networks.
- **Enhanced Detection Reliability:** A major industry challenge in vision-based decentralized monitoring is maintaining accuracy under severe environmental conditions, such as tree shadows, varying illumination, or severe camera perspective distortions. The implementation of the dual-model hybrid decision engine—combining YOLOv8 full-frame localization with MobileNetV2 CNN patch classification—effectively resolves these vulnerabilities, ensuring high-fidelity occupancy data even in structurally challenging outdoor environments.
- **Rapid Deployment and Adaptive Administration:** The integration of the Streamlit web dashboard and the interactive `st_canvas` drawing layer revolutionizes how facility managers configure new sites. Administrators can dynamically upload new camera feeds or custom datasets, manually draw Regions of Interest (ROIs) over the operational layout, and deploy the monitoring system instantly without requiring backend code modifications or advanced technical expertise.
- **Data-Driven Smart City Analytics:** The underlying SQLite database architecture tracks continuous user analysis and automatically logs operational metrics, including hourly occupancy rates and capacity sums. By archiving these analytical logs, parking facility managers and urban planners can extract historical insights, identify peak utilization hours, and optimize urban traffic flow. Ultimately, guiding users to available spaces efficiently reduces localized traffic congestion, fuel consumption, and driver stress.

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ABOUT THE AUTHOR

Sümeyye Buran is a fourth-year Computer Engineering student at Çukurova University. This graduation thesis focuses on the development of a real-time parking occupancy detection system using a hybrid computer vision approach, specifically integrating Ultralytics YOLOv8 object localization with MobileNetV2 Convolutional Neural Network (CNN) patch classification. Alongside deep learning methodologies, her project incorporates modern software engineering practices by developing a full-stack Streamlit web application integrated with dynamic drawing interfaces and an SQLite relational database. The overarching goal of her study is to contribute to smart city infrastructure by providing an efficient, highly scalable, and accessible vision-based parking monitoring solution.

APPENDIX

Appendix A: Smart Parking System Streamlit Web Interface

Otopark Doluluk Tespit Sistemi

[Giriş Yap](#) [Kayıt Ol](#)

Sisteme Giriş

Kullanıcı Adı

Şifre

[Giriş Yap](#)

Otopark Doluluk Tespit Sistemi

[Giriş Yap](#) [Kayıt Ol](#)

Yeni Kullanıcı Kaydı

Kullanıcı Adı Belirleyin

Şifrenizi Belirleyin

Şifrenizi Tekrar Girin

[Hesap Oluştur](#)

Appendix B: Interactive ROI Drawing Canvas via st_canvas

Datasetten Görüntü Seç ve Analiz Et

Tarih seç
2015-11-12

Kamera seç
camera3

Görüntü seç (Dataset içinden)
2015-11-12_0715.jpg

[Analiz Et](#)

Sonuçlar

Kamera
camera3

Toplam Park Yeri
21

Dolu
1

Boş
20

Doluluk Oranı
%4.8

Kamera: camera3 - Görüntü: 2015-11-12_0715.jpg

Hoş geldiniz, user

[Dataset Görüntüsü](#) [Video Simülasyonu](#) [Raporlarım](#) [Harici Dosya Yükle](#)

Kendi Verinizi Ekleyin veya Seçin

Bilgisayarınızdan yeni bir dosya yükleyebilir veya daha önce yüklenip admin tarafından eğitilmiş dataset dosyanızı seçebilirsiniz.

İşlem Tipi:
☐ Bilgisayardan Yeni Yükle ☒ Sistemdeki (Eğitilmiş) Dosyalardan Seç

Dataset İçerisindeki Dosyalar
test.png

☒ Admin tarafından bu dosya için park alanları tanımlanmış! Doğrudan analiz ediliyor.

Analiz Çıktısı

Analiz Tamamlandı: Toplam: 20, Dolu: 13

<https://github.com/sumeyyeburan/Smart-Parking-System>